ECEN 153 HW4 Dylan Fleming

N: $k' = 115\,\mu A/V^2$ $V_T = .43$ $\lambda_n = .06\,V^{-1}$ $\gamma_N \neq 0$ $V_{DSATN} = .63V$

P: $k' = 30\,\mu A/V^2$ $V_T = -.4$ $\lambda_p = -.1\,V^{-1}$ $\gamma_p \neq 0$ $V_{DSATN} = -1V$ $V_{DD} = 2.5V$

1.

a.

$$V_M = \frac{V_{DD} - |V_{TP}| + V_{TN}\sqrt{\frac{B_N}{B_P}}}{1+\sqrt{\frac{B_N}{B_P}}} \qquad B = k'\frac{W}{L}$$

$$1.25 = \frac{2.1 + .43\sqrt{\frac{B_N}{B_P}}}{1+\sqrt{\frac{B_N}{B_P}}}$$

$$1.25 + 1.25\sqrt{\frac{B_N}{B_P}} = 2.1 + .43\sqrt{\frac{B_N}{B_P}}$$

$$.82\sqrt{\frac{B_N}{B_P}} = .85$$

$$\frac{B_N}{B_P} = \left(\frac{.85}{.82}\right)^2$$

$$\frac{115\times10^{-6}\left(\frac{W}{L}\right)_N}{30\times10^{-6}\left(\frac{W}{L}\right)_P} = 1.0745$$

$$\frac{\left(\frac{W}{L}\right)_P}{\left(\frac{W}{L}\right)_N} = \boxed{3.568}$$

b.

$$V_{IH} = V_M - \frac{V_M}{9}, \quad V_{IL} = V_M + \frac{V_{DD}-V_M}{9}$$

$$= 1.25 - \frac{1.25}{-27.5} \qquad = 1.25 + \frac{1.25}{-27.5}$$

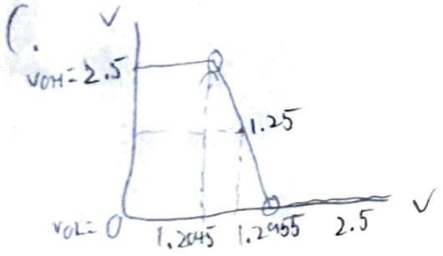
$$= 1.2955V, \qquad = 1.2045V$$

$$NMH = V_{DD} - V_{IH}, \quad NML = V_{IL}$$

$$= 1.2045V \qquad = 1.2045V$$

$$\boxed{NMH = NML = 1.2045V}$$

c.




2.

$$t_p = \frac{.69C_L(R_{eqn} + R_{eqp})}{2}$$

using

$$R_{eq} \approx \frac{3}{4}\frac{V_{DD}}{I_{Dsat}}\left(1 - \frac{7}{9}\times V_{DD}\right)$$

$$R_{eqn} = 13029.979\,\Omega$$

$$R_{eqp} = 13076.740\,\Omega$$

$$I_{DSATN} = 115\times10^{-6}(1)\left((2.5-.43)(.63) - \frac{.63^2}{2}\right)$$

$$= 1.271\times10^{-4}\,A$$

$$I_{DSATP} = 30\times10^{-6}(3.568)\left((2.5-.4)(1) - \frac{1^2}{2}\right)$$

$$= 1.713\times10^{-4}\,A$$

$$\boxed{\begin{aligned} T_{pHL} &= 8.958\times10^{-11}\,s \\ T_{pLH} &= 9.023\times10^{-11}\,s \\ T_p &= 9.005\times10^{-11}\,s \end{aligned}}$$

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HW 4

3. Dynamic Power $= C_L V_{dd}(V_{OH} - V_{OL}) f$

$= 10 \times 10^{-15} (2.5)(2.5)(100 \times 10^{6})$

$= \boxed{6.25 \times 10^{-6}\ W}$